How Syntax Specialization Emerges in Language Models

Introduction. Human sentence processing relies on neural specialization: distinct brain regions and circuits become selectively sensitive to syntactic structure, and this specialization emerges during a limited developmental window often described as a *critical period*. Recent work suggests that large language models (LLMs) show analogous properties, developing neurons, attention heads, and circuits that are selectively responsive to syntax. While such specialization is well documented in trained models, little is known about **how it emerges during training**, **where it localizes**, and **which factors shape its development**. Here we investigate the *developmental dynamics* of syntactic specialization in transformer language models. We ask: **(i)** How does syntactic specialization emerge and evolve over training? **(ii)** How do initialization, model scale, and training data influence this process? To address these questions, we introduce a novel, theory-neutral metric that quantifies internal syntactic specialization without supervised probing.

Methods We introduce the Syntactic Sensitivity Index (SSI), a measure of how consistently internal representations distinguish grammatical from ungrammatical sentences across syntactic phenomena. Using minimal sentence pairs from the BLiMP benchmark (13 syntactic phenomena; >67k sentence pairs), we compute layer-wise and neuron-wise activation differences between grammatical and ungrammatical counterparts. SSI captures the degree to which these contrasts are internally consistent within a phenomenon and distinct across phenomena. We track SSI across training checkpoints for multiple transformer language models, including GPT-2 (124M) and Pythia models, trained from random initialization. Models vary in **random seed**, **training data size** (7GB vs. 13GB), and **model scale** (70M–1.4B parameters). To validate SSI, we relate it to grammaticality-judgment accuracy and conduct **neuron ablation experiments**, comparing targeted ablation of high-SSI neurons to random ablation.

Results Syntactic specialization emerges **gradually** during training: SSI is near zero early on and increases systematically with data exposure. Layer-wise SSI strongly predicts improvements in grammaticality-judgment accuracy ($p < .001$, see figure 1a), outperforming supervised probing methods such as SVMs and regressors. Ablating high-SSI neurons causes substantial increases in sentence perplexity relative to random ablation ($p < .001$), confirming their functional relevance. Across random seeds, models converge on **similar layer-level specialization profiles** despite minimal overlap in the identities of individual syntax-sensitive neurons, indicating convergent functional organization with divergent micro-implementations. Crucially, specialization trajectories diverge early in training but converge after approximately **16M tokens**, revealing a **critical period of syntactic specialization** analogous to early neural plasticity in human language acquisition (see figure 1b). Model scale amplifies syntactic specialization, yielding sharper and more localized SSI peaks (see figure 2), while training data size interacts with linguistic phenomenon: regular dependencies (e.g., agreement) benefit from more data, whereas irregular constructions show more distributed representations.

Conclusion These findings provide a mechanistic account of **when, where, and how** syntactic specialization emerges in language models. By introducing SSI, we show that syntactic knowledge is not pre-wired in transformer architectures but develops through training, follows a staged trajectory with an early critical period, and converges across models despite neuron-level variability. The parallels between model development and human language acquisition suggest that training dynamics, not just architecture, play a central role in shaping syntactic representations. This work offers new tools and perspectives for connecting computational models with theories of sentence processing and language development.

Reference

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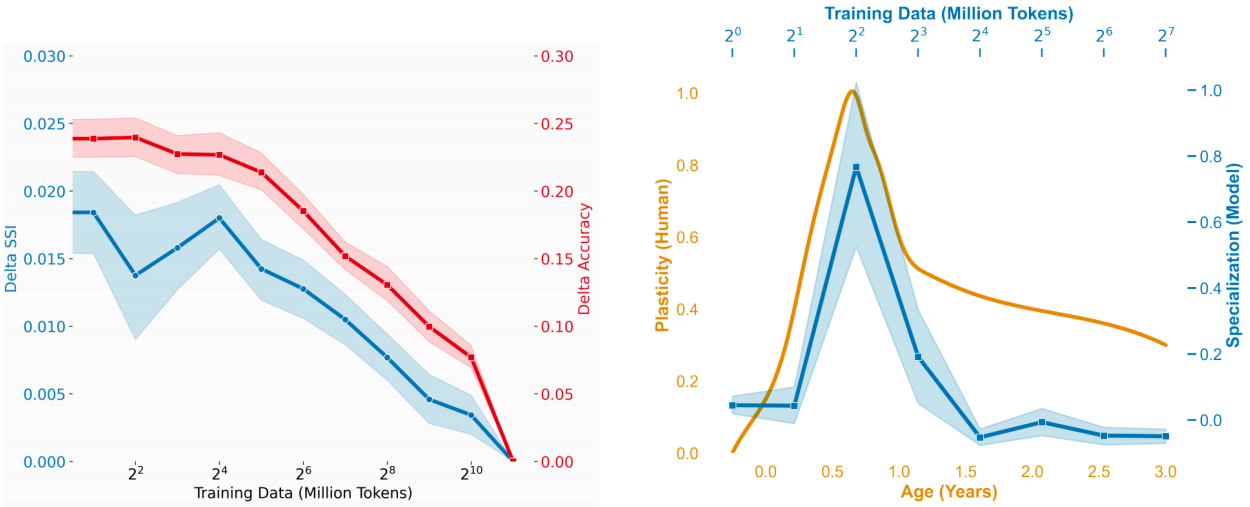

Figure 1. (a) Left. Differences in SSI and grammatical judgment task accuracy between various checkpoints during training and the final model. The correlated downward trends suggest that increased syntactic specialization (lower ΔSSI) is associated with improved grammatical accuracy (lower ΔAccuracy). (b) Right. Human neural plasticity (orange) peaks early in development, consistent with the critical period for language acquisition (Keshavan et al., 2014). Analogously, model syntactic specialization (blue), measured by SSI divergence across seeds, emerges early in training and converges after approximately 16 million tokens

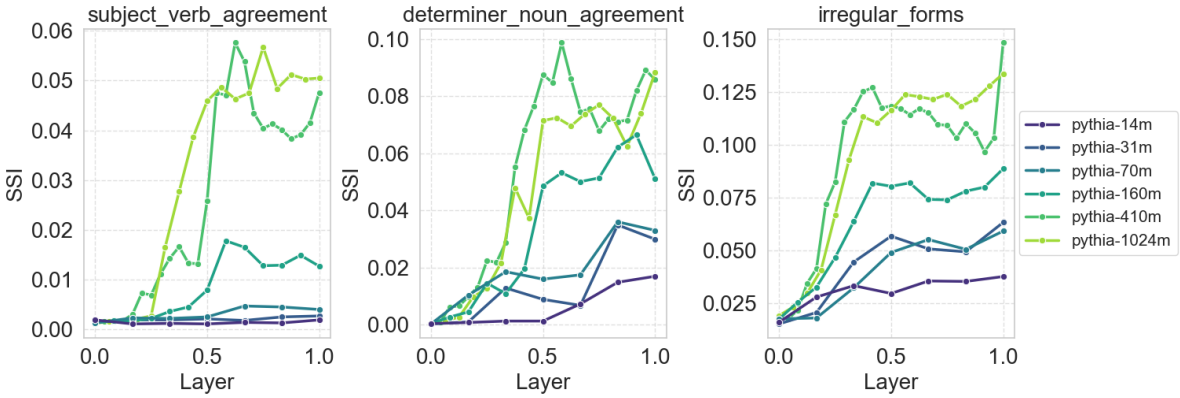

Figure 2. Model scale analysis. Each line represents a model of a certain parameter size in the Pythia series. Layer is normalized for comparison.